

■ 시나리오 별 백업 복구(User Managed) - Archive log mode

KEY WORDS

- <시나리오 20> Redo log file(current) 삭제된 후 DB 정상적인 종료
- <시나리오 21> Redo log file(current) 삭제된 후 DB 비정상적인 종료(일관성 없는 백업, online, hot)
- <시나리오 22> 운영 중에 Redo log file(current) 삭제
- <시나리오 22.1> 22 번 시나리오를 clear unarchived 로 복구
- <시나리오 23> DB 정상적인 종료 후에 controlfile 손상시 복구
- <시나리오 24> DB 정상적인 종료 후에 controlfile 손상시 복구(컨트롤파일 재생성)
- <시나리오 25> DB 비정상적인 종료 후에 controlfile 손상시 복구(컨트롤파일 재생성)

<<시나리오 20>> current redo log file 삭제된 후 DB 정상적인 종료

```
SYS@orcl> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status, b.archived, b.first_change#,
b.next_change# from v$logfile a, v$log b where a.group# = b.group# order by 1,2
```

GROUP#	SEQUENCE#	MEMBER	MB	STATUS	ARC	FIRST_CHANGE#	NEXT_CHANGE#
1	10	/u01/app/oracle/oradata/ORCL/redo01.log	10	CURRENT	NO		
2	8	/u01/app/oracle/oradata/ORCL/redo02.log	10	INACTIVE	YES		
3	9	/u01/app/oracle/oradata/ORCL/redo03.log	10	INACTIVE	YES		

```
SYS@orcl> select sequence#, name, FIRST_CHANGE#, NEXT_CHANGE# from v$archived_log
```

SEQUENCE#	NAME	FIRST_CHANGE#	NEXT_CHANGE#
54	/home/oracle/arch2/arch_1_54_1200525254.arc	2886751	2888319
55	/home/oracle/arch1/arch_1_55_1200525254.arc	2888319	2890206
55	/home/oracle/arch2/arch_1_55_1200525254.arc	2888319	2890206
55	/home/oracle/arch1/arch_1_55_1200525254.arc	2888319	2890206
55	/home/oracle/arch2/arch_1_55_1200525254.arc	2888319	2890206
54	/home/oracle/arch1/arch_1_54_1200525254.arc	2886751	2888319
54	/home/oracle/arch2/arch_1_54_1200525254.arc	2886751	2888319
1	/home/oracle/arch1/arch_1_1_1201040433.arc	2897913	2997921
1	/home/oracle/arch2/arch_1_1_1201040433.arc	2897913	2997921
2	/home/oracle/arch1/arch_1_2_1201040433.arc	2997921	3097929

```
2 /home/oracle/arch2/arch_1_2_1201040433.arc          2997921      3097929
```

SEQUENCE#	NAME	FIRST_CHANGE#	NEXT_CHANGE#
3	/home/oracle/arch1/arch_1_3_1201040433.arc	3097929	3099666
3	/home/oracle/arch2/arch_1_3_1201040433.arc	3097929	3099666
4	/home/oracle/arch1/arch_1_4_1201040433.arc	3099666	3099798
4	/home/oracle/arch2/arch_1_4_1201040433.arc	3099666	3099798
5	/home/oracle/arch1/arch_1_5_1201040433.arc	3099798	3100660
5	/home/oracle/arch2/arch_1_5_1201040433.arc	3099798	3100660
6	/home/oracle/arch1/arch_1_6_1201040433.arc	3100660	3101928
6	/home/oracle/arch2/arch_1_6_1201040433.arc	3100660	3101928
7	/home/oracle/arch1/arch_1_7_1201040433.arc	3101928	3103597
7	/home/oracle/arch2/arch_1_7_1201040433.arc	3101928	3103597
8	/home/oracle/arch1/arch_1_8_1201040433.arc	3103597	3106279
8	/home/oracle/arch2/arch_1_8_1201040433.arc	3103597	3106279
9	/home/oracle/arch1/arch_1_9_1201040433.arc	3106279	3106288
9	/home/oracle/arch2/arch_1_9_1201040433.arc	3106279	3106288

80 rows selected.

```
SYS@orcl> select a.file#, a.name, a.checkpoint_change#, b.status, b.change#, to_char(b.time, 'yyyy-mm-dd
hh24:mi:ss.sssss') time from v$datafile a, v$backup b where a.file# = b.file#;
```

FILE#	NAME	CHECKPOINT_CHANGE#	STATUS
CHANGE#	TIME		
1	/u01/app/oracle/oradata/ORCL/system01.dbf	3106618	NOT ACTIVE
3106208	2025-05-13 22:34:25.81265		
3	/u01/app/oracle/oradata/ORCL/sysaux01.dbf	3106618	NOT ACTIVE
3106208	2025-05-13 22:34:25.81265		
4	/u01/app/oracle/oradata/ORCL/undotbs01.dbf	3106618	NOT ACTIVE
3106208	2025-05-13 22:34:25.81265		
7	/u01/app/oracle/oradata/ORCL/users01.dbf	3106618	NOT ACTIVE
3106208	2025-05-13 22:34:25.81265		

```
SYS@orcl> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status, b.archived, b.first_change#,
b.next_change# from v$logfile a, v$log b where a.group# = b.group# order by 1,2;
```

GROUP#	SEQUENCE#	MEMBER	MB	STATUS
ARC	FIRST_CHANGE#	NEXT_CHANGE#		

1	10	/u01/app/oracle/oradata/ORCL/redo01.log	10	CURRENT
3106288	1.8447E+19			NO
2	8	/u01/app/oracle/oradata/ORCL/redo02.log	10	INACTIVE
3103597	3106279			YES
3	9	/u01/app/oracle/oradata/ORCL/redo03.log	10	INACTIVE
3106279	3106288			YES

```
SYS@orcl> alter system switch logfile;
System altered.
```

```
SYS@orcl> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status, b.archived, b.first_change#,
b.next_change# from v$logfile a, v$log b where a.group# = b.group# order by 1,2;
```

GROUP#	SEQUENCE#	MEMBER	MB	STATUS
ARC	FIRST_CHANGE#	NEXT_CHANGE#		

1	10	/u01/app/oracle/oradata/ORCL/redo01.log	10	ACTIVE
3106288	3107564			YES
2	11	/u01/app/oracle/oradata/ORCL/redo02.log	10	CURRENT
3107564	1.8447E+19			NO
3	9	/u01/app/oracle/oradata/ORCL/redo03.log	10	INACTIVE
3106279	3106288			YES

```
-- Alert log file
2025-05-13T23:02:17.634117+09:00
Thread 1 advanced to log sequence 11 (LGWR switch)
Current log# 2 seq# 11 mem# 0: /u01/app/oracle/oradata/ORCL/redo02.log
```

```
# Table 생성
```

```
SYS@orcl> create table hr.emp1 tablespace users as select * from hr.employees;
Table created.
```

```
SYS@orcl> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status, b.archived, b.first_change#,
```

b.next_change# from v\$logfile a, v\$log b where a.group# = b.group# order by 1,2;

GROUP#	SEQUENCE#	MEMBER	MB	STATUS
ARC	FIRST_CHANGE#	NEXT_CHANGE#		

1	10	/u01/app/oracle/oradata/ORCL/redo01.log	10	INACTIVE
3106288	3107564			YES
2	11	/u01/app/oracle/oradata/ORCL/redo02.log	10	CURRENT
3107564	1.8447E+19			NO
3	9	/u01/app/oracle/oradata/ORCL/redo03.log	10	INACTIVE
3106279	3106288			YES

장애 발생

```
SYS@orcl> ! ls /u01/app/oracle/oradata/ORCL/redo02.log
/u01/app/oracle/oradata/ORCL/redo02.log
```

```
SYS@orcl> ! rm /u01/app/oracle/oradata/ORCL/redo02.log
```

```
SYS@orcl> ! ls /u01/app/oracle/oradata/ORCL/redo02.log
ls: cannot access /u01/app/oracle/oradata/ORCL/redo02.log: No such file or directory
```

정상적인 종료

```
SYS@orcl> shutdown immediate
Database closed.
Database dismounted.
ORACLE instance shut down.
```

```
-- Alert log File(문제 없음)
2025-05-13T23:08:13.408102+09:00
Instance shutdown complete (OS id: 22085)
```

```
SYS@orcl> startup
ORACLE instance started.
```

```
Total System Global Area 1543500144 bytes
Fixed Size                  8896880 bytes
Variable Size              905969664 bytes
Database Buffers          620756992 bytes
```

Redo Buffers 7876608 bytes

Database mounted.

ORA-03113: end-of-file on communication channel

Process ID: 23316

Session ID: 33 Serial number: 46100

-- Alert log File

2025-05-13T23:09:26.499194+09:00

Errors in file /u01/app/oracle/diag/rdbms/orcl/orcl/trace/orcl_lgwr_23276.trc:

ORA-00313: open failed for members of log group 2 of thread 1

ORA-00312: online log 2 thread 1: '/u01/app/oracle/oradata/ORCL/redo02.log'

ORA-27037: unable to obtain file status

Linux-x86_64 Error: 2: No such file or directory

Additional information: 7

SYS@orcl> conn / as sysdba

Connected to an idle instance.

SYS@orcl> ! ls /u01/app/oracle/oradata/ORCL/*.log

/u01/app/oracle/oradata/ORCL/redo01.log /u01/app/oracle/oradata/ORCL/redo03.log

SYS@orcl> startup mount

ORACLE instance started.

Total System Global Area 1543500144 bytes

Fixed Size 8896880 bytes

Variable Size 905969664 bytes

Database Buffers 620756992 bytes

Redo Buffers 7876608 bytes

Database mounted.

SYS@orcl> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status, b.archived, b.first_change#, b.next_change# from v\$logfile a, v\$log b where a.group# = b.group# order by 1,2;

GROUP#	SEQUENCE#	MEMBER	MB	STATUS	ARC FIRST_CHANGE#	NEXT_CHANGE#
1	10	/u01/app/oracle/oradata/ORCL/redo01.log	10	INACTIVE	YES	
3106288	3107564					
2	11	/u01/app/oracle/oradata/ORCL/redo02.log	10	CURRENT	NO	

```

3107564 1.8447E+19
          3          9 /u01/app/oracle/oradata/ORCL/redo03.log          10 INACTIVE          YES
3106279    3106288

```

```

SYS@orcl> recover database until cancel
Media recovery complete.

```

```

SYS@orcl> ! ls /u01/app/oracle/oradata/ORCL/*.log
/u01/app/oracle/oradata/ORCL/redo01.log  /u01/app/oracle/oradata/ORCL/redo03.log

```

```

SYS@orcl> alter database open resetlogs;
Database altered.

```

```

SYS@orcl> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status, b.archived, b.first_change#,
b.next_change# from v$logfile a, v$log b where a.group# = b.group# order by 1,2;

```

GROUP#	SEQUENCE#	MEMBER	MB	STATUS
ARC	FIRST_CHANGE#	NEXT_CHANGE#		
1	1	/u01/app/oracle/oradata/ORCL/redo01.log	10	CURRENT
2	0	/u01/app/oracle/oradata/ORCL/redo02.log	10	UNUSED
0	0			
3	0	/u01/app/oracle/oradata/ORCL/redo03.log	10	UNUSED
0	0			

```

SYS@orcl> !ls /u01/app/oracle/oradata/ORCL/
control01.ctl control_20250513.ctl initorcl.ora redo02.log sysaux01.dbf temp01.dbf users01.dbf
control02.ctl control_2025051301.ctl redo01.log redo03.log system01.dbf undotbs01.dbf

```

```

SYS@orcl> select count(*) from hr.emp1;

```

```

COUNT(*)
-----
107

```

```

SYS@orcl> select * from v$database_incarnation;

```

```

INCARNATION# RESETLOGS_CHANGE# RESETLOGS_TIME      PRIOR_RESETLOGS_CHANGE#
PRIOR_RESETLOGS_TI  STATUS      RESETLOGS_ID  PRIOR_INCARNATION#  FLASHBACK_DATABASE_ALLOWED
CON_ID
-----

```

```

          1              1 17-APR-19              0              PARENT
1005785759          0 NO              0
          2          1920977 01-MAY-25              1 17-APR-19          PARENT
1199933387          1 NO              0
          3          2310250 07-MAY-25          1920977 01-MAY-25          PARENT
1200525254          2 NO              0
          4          2897913 13-MAY-25          2310250 07-MAY-25          PARENT
1201040433          3 NO              0
          5          3109655 13-MAY-25          2897913 13-MAY-25          CURRENT
1201043631          4 NO
          3 NO              0

```

온라인 백업

```
SYS@orcl> ! rm /home/oracle/backup/open_bkp/*
```

```
SYS@orcl> alter database begin backup;
```

Database altered.

```

SYS@orcl> ! cp -v /u01/app/oracle/oradata/ORCL/*.dbf /home/oracle/backup/open_bkp/
'/u01/app/oracle/oradata/ORCL/sysaux01.dbf' -> '/home/oracle/backup/open_bkp/sysaux01.dbf'
'/u01/app/oracle/oradata/ORCL/system01.dbf' -> '/home/oracle/backup/open_bkp/system01.dbf'
'/u01/app/oracle/oradata/ORCL/temp01.dbf' -> '/home/oracle/backup/open_bkp/temp01.dbf'
'/u01/app/oracle/oradata/ORCL/undotbs01.dbf' -> '/home/oracle/backup/open_bkp/undotbs01.dbf'
'/u01/app/oracle/oradata/ORCL/users01.dbf' -> '/home/oracle/backup/open_bkp/users01.dbf'

```

```
SYS@orcl> alter database end backup;
```

Database altered.

```

SYS@orcl> select a.file#, a.name, a.checkpoint_change#, b.status, b.change#, to_char(b.time, 'yyyy-mm-dd
hh24:mi:ss.sssss') time from v$datafile a, v$backup b where a.file# = b.file#;

```

```

FILE# NAME                                CHECKPOINT_CHANGE# STATUS
CHANGE# TIME

```

```

-----
1 /u01/app/oracle/oradata/ORCL/system01.dbf          3110626 NOT ACTIVE
3110626 2025-05-13 23:18:54.83934
3 /u01/app/oracle/oradata/ORCL/sysaux01.dbf          3110626 NOT ACTIVE
3110626 2025-05-13 23:18:54.83934
4 /u01/app/oracle/oradata/ORCL/undotbs01.dbf          3110626 NOT ACTIVE
3110626 2025-05-13 23:18:54.83934
7 /u01/app/oracle/oradata/ORCL/users01.dbf            3110626 NOT ACTIVE
3110626 2025-05-13 23:18:54.83934

```

SYS@orcl> alter database backup controlfile to '/home/oracle/backup/open_bkp/control01.ctl';

Database altered.

SYS@orcl> alter system archive log current;

System altered.

SYS@orcl> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status, b.archived, b.first_change#, b.next_change# from v\$logfile a, v\$log b where a.group# = b.group# order by 1,2;

GROUP#	SEQUENCE#	MEMBER	MB	STATUS
ARC	FIRST_CHANGE#	NEXT_CHANGE#		
1	1	/u01/app/oracle/oradata/ORCL/redo01.log	10	ACTIVE
3109655	3110708			YES
2	2	/u01/app/oracle/oradata/ORCL/redo02.log	10	CURRENT
3110708	1.8447E+19			NO
3	0	/u01/app/oracle/oradata/ORCL/redo03.log	10	UNUSED
	0	0		YES

SYS@orcl> select a.file#, a.name, a.checkpoint_change#, b.status, b.change#, to_char(b.time, 'yyyy-mm-dd hh24:mi:ss.sssss') time from v\$datafile a, v\$backup b where a.file# = b.file#;

FILE#	NAME	CHECKPOINT_CHANGE#	STATUS
CHANGE#	TIME		


```

1 /u01/app/oracle/oradata/ORCL/system01.dbf          3110626 NOT ACTIVE
3110626 2025-05-13 23:18:54.83934
3 /u01/app/oracle/oradata/ORCL/sysaux01.dbf          3110626 NOT ACTIVE
3110626 2025-05-13 23:18:54.83934
4 /u01/app/oracle/oradata/ORCL/undotbs01.dbf         3110626 NOT ACTIVE
3110626 2025-05-13 23:18:54.83934
7 /u01/app/oracle/oradata/ORCL/users01.dbf           3110626 NOT ACTIVE
3110626 2025-05-13 23:18:54.83934

```

SYS@orcl> select sequence#, name, FIRST_CHANGE#, NEXT_CHANGE# from v\$archived_log;

SEQUENCE#	NAME	FIRST_CHANGE#	NEXT_CHANGE#
24	/home/oracle/archlog/arch2/arch_1_24_1200525254.arc	2339146	2341425
25	/home/oracle/archlog/arch2/arch_1_25_1200525254.arc	2341425	2341453
26	/home/oracle/archlog/arch2/arch_1_26_1200525254.arc	2341453	2341456
27	/home/oracle/archlog/arch2/arch_1_27_1200525254.arc	2341456	2341619
28	/home/oracle/archlog/arch2/arch_1_28_1200525254.arc	2341619	2345364
29	/home/oracle/archlog/arch2/arch_1_29_1200525254.arc	2345364	2347853
30	/home/oracle/arch1/arch_1_30_1200525254.arc	2347853	2354351
30	/home/oracle/arch2/arch_1_30_1200525254.arc	2347853	2354351
31	/home/oracle/arch1/arch_1_31_1200525254.arc	2354351	2356143
31	/home/oracle/arch2/arch_1_31_1200525254.arc	2354351	2356143
32	/home/oracle/arch1/arch_1_32_1200525254.arc	2356143	2456244

SEQUENCE#	NAME	FIRST_CHANGE#	NEXT_CHANGE#
32	/home/oracle/arch2/arch_1_32_1200525254.arc	2356143	2456244
33	/home/oracle/arch1/arch_1_33_1200525254.arc	2456244	2457117
33	/home/oracle/arch2/arch_1_33_1200525254.arc	2456244	2457117
34	/home/oracle/arch1/arch_1_34_1200525254.arc	2457117	2459505
34	/home/oracle/arch2/arch_1_34_1200525254.arc	2457117	2459505
35	/home/oracle/arch1/arch_1_35_1200525254.arc	2459505	2463463
35	/home/oracle/arch2/arch_1_35_1200525254.arc	2459505	2463463
36	/home/oracle/arch1/arch_1_36_1200525254.arc	2463463	2463466
36	/home/oracle/arch2/arch_1_36_1200525254.arc	2463463	2463466
37	/home/oracle/arch1/arch_1_37_1200525254.arc	2463466	2463469
37	/home/oracle/arch2/arch_1_37_1200525254.arc	2463466	2463469

SEQUENCE#	NAME	FIRST_CHANGE#	NEXT_CHANGE#
-----------	------	---------------	--------------

38 /home/oracle/arch1/arch_1_38_1200525254.arc	2463469	2463472
38 /home/oracle/arch2/arch_1_38_1200525254.arc	2463469	2463472
39 /home/oracle/arch1/arch_1_39_1200525254.arc	2463472	2468896
39 /home/oracle/arch2/arch_1_39_1200525254.arc	2463472	2468896
40 /home/oracle/arch1/arch_1_40_1200525254.arc	2468896	2468908
40 /home/oracle/arch2/arch_1_40_1200525254.arc	2468896	2468908
41 /home/oracle/arch1/arch_1_41_1200525254.arc	2468908	2468912
41 /home/oracle/arch2/arch_1_41_1200525254.arc	2468908	2468912
42 /home/oracle/arch1/arch_1_42_1200525254.arc	2468912	2468983
42 /home/oracle/arch2/arch_1_42_1200525254.arc	2468912	2468983
43 /home/oracle/arch1/arch_1_43_1200525254.arc	2468983	2569464

SEQUENCE#	NAME	FIRST_CHANGE#	NEXT_CHANGE#
-----------	------	---------------	--------------

3	/home/oracle/arch1/arch_1_3_1201040433.arc	3097929	3099666
3	/home/oracle/arch2/arch_1_3_1201040433.arc	3097929	3099666
4	/home/oracle/arch1/arch_1_4_1201040433.arc	3099666	3099798
4	/home/oracle/arch2/arch_1_4_1201040433.arc	3099666	3099798
5	/home/oracle/arch1/arch_1_5_1201040433.arc	3099798	3100660
5	/home/oracle/arch2/arch_1_5_1201040433.arc	3099798	3100660
6	/home/oracle/arch1/arch_1_6_1201040433.arc	3100660	3101928
6	/home/oracle/arch2/arch_1_6_1201040433.arc	3100660	3101928
7	/home/oracle/arch1/arch_1_7_1201040433.arc	3101928	3103597
7	/home/oracle/arch2/arch_1_7_1201040433.arc	3101928	3103597
8	/home/oracle/arch1/arch_1_8_1201040433.arc	3103597	3106279

SEQUENCE#	NAME	FIRST_CHANGE#	NEXT_CHANGE#
-----------	------	---------------	--------------

8	/home/oracle/arch2/arch_1_8_1201040433.arc	3103597	3106279
9	/home/oracle/arch1/arch_1_9_1201040433.arc	3106279	3106288
9	/home/oracle/arch2/arch_1_9_1201040433.arc	3106279	3106288
10	/home/oracle/arch1/arch_1_10_1201040433.arc	3106288	3107564
10	/home/oracle/arch2/arch_1_10_1201040433.arc	3106288	3107564
1	/home/oracle/arch1/arch_1_1_1201043631.arc	3109655	3110708
1	/home/oracle/arch2/arch_1_1_1201043631.arc	3109655	3110708

<<시나리오21>> current redo log file 삭제된 후 DB 비정상적인 종료(일관성 없는 백업,

online, hot, open)

```
SYS@orcl> alter system switch logfile;
```

System altered.

```
SYS@orcl> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status, b.archived, b.first_change#,
b.next_change# from v$logfile a, v$log b where a.group# = b.group# order by 1,2;
```

GROUP#	SEQUENCE#	MEMBER	MB	STATUS	
ARC	FIRST_CHANGE#	NEXT_CHANGE#			

1	1	/u01/app/oracle/oradata/ORCL/redo01.log	10	ACTIVE	YES
3109655	3110708				
2	2	/u01/app/oracle/oradata/ORCL/redo02.log	10	ACTIVE	YES
3110708	3110790				
3	3	/u01/app/oracle/oradata/ORCL/redo03.log	10	CURRENT	NO
3110790	1.8447E+19				

```
SYS@orcl> create table hr.dept1 tablespace users as select * from hr.departments;
```

Table created.

```
SYS@orcl> select count(*) from hr.dept1;
```

```

COUNT(*)
-----
27
```

장애 발생

```
SYS@orcl> ! ls /u01/app/oracle/oradata/ORCL/redo03.log
/u01/app/oracle/oradata/ORCL/redo03.log
```

```
SYS@orcl> ! rm /u01/app/oracle/oradata/ORCL/redo03.log
```

```
SYS@orcl> ! ls /u01/app/oracle/oradata/ORCL/redo03.log
```

ls: cannot access /u01/app/oracle/oradata/ORCL/redo03.log: No such file or directory

DB 비정상적인 종료

```
SYS@orcl> shutdown abort
ORACLE instance shut down.
```

```
-- Alert log file
2025-05-13T23:27:46.666261+09:00
Shutting down ORACLE instance (abort) (OS id: 23563)
License high water mark = 3
2025-05-13T23:27:46.667088+09:00
USER (ospid: 23563): terminating the instance
2025-05-13T23:27:47.793425+09:00
Instance terminated by USER, pid = 23563
2025-05-13T23:27:47.887580+09:00
Instance shutdown complete (OS id: 23563)
```

오류확인

```
SYS@orcl> startup
ORACLE instance started.
```

```
Total System Global Area 1543500144 bytes
Fixed Size                  8896880 bytes
Variable Size              905969664 bytes
Database Buffers           620756992 bytes
Redo Buffers                7876608 bytes
Database mounted.
ORA-00313: open failed for members of log group 3 of thread 1
ORA-00312: online log 3 thread 1: '/u01/app/oracle/oradata/ORCL/redo03.log'
ORA-27037: unable to obtain file status
Linux-x86_64 Error: 2: No such file or directory
Additional information: 7
```

```
-- Alert log file
2025-05-13T23:29:25.840396+09:00
Beginning crash recovery of 1 threads
  Thread 1: Recovery starting at checkpoint rba (logseq 2 block 4), scn 0
```

```
-- Instance Recovery
22025-05-13T23:29:25.855460+09:00
Started redo scan
2025-05-13T23:29:25.856401+09:00
Errors in file /u01/app/oracle/diag/rdbms/orcl/orcl/trace/orcl_ora_24892.trc:
ORA-00313: open failed for members of log group 3 of thread 1
```

ORA-00312: online log 3 thread 1: '/u01/app/oracle/oradata/ORCL/redo03.log'

ORA-27037: unable to obtain file status

Linux-x86_64 Error: 2: No such file or directory

Additional information: 7

SYS@orcl> select status from v\$instance;

STATUS

MOUNTED

SYS@orcl> select a.file#, b.name tbs_name, a.name file_name, a.checkpoint_change#, a.status from v\$datafile a,
v\$tablespace b where a.ts# = b.ts#;

FILE# TBS_NAME

FILE_NAME

CHECKPOINT_CHANGE# STATUS

1 SYSTEM

/u01/app/oracle/oradata/ORCL/system01.dbf

3110708 SYSTEM

3 SYSAUX

/u01/app/oracle/oradata/ORCL/sysaux01.dbf

3110708 ONLINE

FILE# TBS_NAME

FILE_NAME

CHECKPOINT_CHANGE# STATUS

4 UNDOTBS1

/u01/app/oracle/oradata/ORCL/undotbs01.dbf

3110708 ONLINE

7 USERS

/u01/app/oracle/oradata/ORCL/users01.dbf

FILE# TBS_NAME

FILE_NAME

CHECKPOINT_CHANGE# STATUS

3110708 ONLINE

```
SYS@orcl> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status, b.archived, b.first_change#,
b.next_change# from v$logfile a, v$log b where a.group# = b.group# order by 1,2;
```

GROUP#	SEQUENCE#	MEMBER	MB	STATUS	
ARC	FIRST_CHANGE#	NEXT_CHANGE#			
1	1	/u01/app/oracle/oradata/ORCL/redo01.log	10	INACTIVE	YES
3109655	3110708				
2	2	/u01/app/oracle/oradata/ORCL/redo02.log	10	ACTIVE	YES
3110708	3110790				
3	3	/u01/app/oracle/oradata/ORCL/redo03.log	10	CURRENT	NO
3110790	1.8447E+19				

Recovery

```
SYS@orcl> ! cp -v /home/oracle/backup/open_bkp/*.dbf /u01/app/oracle/oradata/ORCL/
'/home/oracle/backup/open_bkp/sysaux01.dbf' -> '/u01/app/oracle/oradata/ORCL/sysaux01.dbf'
'/home/oracle/backup/open_bkp/system01.dbf' -> '/u01/app/oracle/oradata/ORCL/system01.dbf'
'/home/oracle/backup/open_bkp/temp01.dbf' -> '/u01/app/oracle/oradata/ORCL/temp01.dbf'
'/home/oracle/backup/open_bkp/undotbs01.dbf' -> '/u01/app/oracle/oradata/ORCL/undotbs01.dbf'
'/home/oracle/backup/open_bkp/users01.dbf' -> '/u01/app/oracle/oradata/ORCL/users01.dbf'
```

```
SYS@orcl> recover database until cancel
```

ORA-00279: change 3110626 generated at 05/13/2025 23:18:54 needed for thread 1

ORA-00289: suggestion : /home/oracle/arch2/arch_1_1_1201043631.arc

ORA-00280: change 3110626 for thread 1 is in sequence #1

Specify log: {<RET>=suggested | filename | AUTO | CANCEL}

ORA-00279: change 3110708 generated at 05/13/2025 23:21:24 needed for thread 1
ORA-00289: suggestion : /home/oracle/arch2/arch_1_2_1201043631.arc
ORA-00280: change 3110708 for thread 1 is in sequence #2
ORA-00278: log file '/home/oracle/arch2/arch_1_1_1201043631.arc' no longer needed for this recovery

Specify log: {<RET>=suggested | filename | AUTO | CANCEL}

ORA-00279: change 3110790 generated at 05/13/2025 23:24:13 needed for thread 1
ORA-00289: suggestion : /home/oracle/arch2/arch_1_3_1201043631.arc
ORA-00280: change 3110790 for thread 1 is in sequence #3
ORA-00278: log file '/home/oracle/arch2/arch_1_2_1201043631.arc' no longer needed for this recovery

Specify log: {<RET>=suggested | filename | AUTO | CANCEL}

ORA-00308: cannot open archived log '/home/oracle/arch2/arch_1_3_1201043631.arc'
ORA-27037: unable to obtain file status
Linux-x86_64 Error: 2: No such file or directory
Additional information: 7

SYS@orcl> recover database until cancel

ORA-00279: change 3110790 generated at 05/13/2025 23:24:13 needed for thread 1
ORA-00289: suggestion : /home/oracle/arch2/arch_1_3_1201043631.arc
ORA-00280: change 3110790 for thread 1 is in sequence #3

Specify log: {<RET>=suggested | filename | AUTO | CANCEL}
cancel
Media recovery cancelled.

SYS@orcl> alter database open resetlogs;

Database altered.

===== cancel base recovery 실패한 경우 =====

create pfile from spfile;

shutdown abort

! vi \$ORACLE_HOME/dbs/initiorcl.ora

-- 추가

_allow_resetlogs_corruption=true

startup pfile=\$ORACLE_HOME/dbs/initiorcl.ora mount

alter database open resetlogs;

온라인 백업(sh onbkp.sh 수행)

SYS@orcl> ! rm /home/oracle/backup/open_bkp/*

SYS@orcl> alter database begin backup;

Database altered.

SYS@orcl> ! cp -v /u01/app/oracle/oradata/ORCL/*.dbf /home/oracle/backup/open_bkp/

SYS@orcl> alter database end backup;

Database altered.

SYS@orcl> select a.file#, a.name, a.checkpoint_change#, b.status, b.change#, to_char(b.time, 'yyyy-mm-dd hh24:mi:ss.sssss') time from v\$datafile a, v\$backup b where a.file# = b.file#;

SYS@orcl> alter database backup controlfile to '/home/oracle/backup/open_bkp/control01.ctl';

Database altered.

SYS@orcl> alter system archive log current;

System altered.


```
SYS@orcl> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status, b.archived, b.first_change#,  
b.next_change# from v$logfile a, v$log b where a.group# = b.group# order by 1,2;
```

```
SYS@orcl> select a.file#, a.name, a.checkpoint_change#, b.status, b.change#, to_char(b.time, 'yyyy-mm-dd  
hh24:mi:ss.sssss') time from v$datafile a, v$backup b where a.file# = b.file#;
```

```
SYS@orcl> select sequence#, name, FIRST_CHANGE#, NEXT_CHANGE# from v$archived_log;
```

<<시나리오 22>> 운영중에 current redo log file 삭제

```
SYS@orcl> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status, b.archived, b.first_change#,
b.next_change# from v$logfile a, v$log b where a.group# = b.group# order by 1,2;
```

```
GROUP# SEQUENCE#
-----
MEMBER
-----
MB STATUS          ARC FIRST_CHANGE# NEXT_CHANGE#
-----
1          4
/u01/app/oracle/oradata/ORCL/redo01.log
10 INACTIVE        YES          3113243      3113334

2          5
/u01/app/oracle/oradata/ORCL/redo02.log
10 INACTIVE        YES          3113334      3113469
```

```
GROUP# SEQUENCE#
-----
MEMBER
-----
MB STATUS          ARC FIRST_CHANGE# NEXT_CHANGE#
-----
3          6
/u01/app/oracle/oradata/ORCL/redo03.log
10 CURRENT         NO           3113469      1.8447E+19
```

```
SYS@orcl> alter system switch logfile;
```

System altered.

장애 발생

```
SYS@orcl> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status, b.archived, b.first_change#,
b.next_change# from v$logfile a, v$log b where a.group# = b.group# order by 1,2;
```

```
GROUP# SEQUENCE#
```

MEMBER

MB STATUS ARC FIRST_CHANGE# NEXT_CHANGE#

1 4
/u01/app/oracle/oradata/ORCL/redo01.log
10 INACTIVE YES 3113243 3113334

2 5
/u01/app/oracle/oradata/ORCL/redo02.log
10 INACTIVE YES 3113334 3113469

GROUP# SEQUENCE#

MEMBER

MB STATUS ARC FIRST_CHANGE# NEXT_CHANGE#

3 6
/u01/app/oracle/oradata/ORCL/redo03.log
10 CURRENT NO 3113469 1.8447E+19

SYS@orcl> ! rm /u01/app/oracle/oradata/ORCL/redo03.log

SYS@orcl> alter system switch logfile;
System altered.

-- Alert log file 확인

2025-05-13T23:29:25.856401+09:00

Errors in file /u01/app/oracle/diag/rdbms/orcl/orcl/trace/orcl_ora_24892.trc:

ORA-00313: open failed for members of log group 3 of thread 1

ORA-00312: online log 3 thread 1: '/u01/app/oracle/oradata/ORCL/redo03.log'

ORA-27037: unable to obtain file status

Linux-x86_64 Error: 2: No such file or directory

Additional information: 7

Recovery

SYS@orcl> alter database clear unarchived logfile group 3;

Database altered.

SYS@orcl> ! ls /u01/app/oracle/oradata/ORCL/*.log

/u01/app/oracle/oradata/ORCL/redo01.log

/u01/app/oracle/oradata/ORCL/redo02.log

/u01/app/oracle/oradata/ORCL/redo03.log

SYS@orcl> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status, b.archived, b.first_change#, b.next_change# from v\$logfile a, v\$log b where a.group# = b.group# order by 1,2;

GROUP#	SEQUENCE#	MEMBER	MB	STATUS	ARC	FIRST_CHANGE#	NEXT_CHANGE#
1	7	/u01/app/oracle/oradata/ORCL/redo01.log	10	INACTIVE	YES	3114394	3114461
2	8	/u01/app/oracle/oradata/ORCL/redo02.log	10	CURRENT	NO	3114461	1.8447E+19

GROUP#	SEQUENCE#	MEMBER	MB	STATUS	ARC	FIRST_CHANGE#	NEXT_CHANGE#
3	0	/u01/app/oracle/oradata/ORCL/redo03.log	10	UNUSED	YES	3113469	3114394

SYS@orcl> select sequence#, name, FIRST_CHANGE#, NEXT_CHANGE# from v\$archived_log;

=====## Inactive 상태의 로그파일 손상시=====

```
SYS@orcl> ! rm /u01/app/oracle/oradata/orcl/redo01.log
```

```
SYS@orcl> alter database clear logfile group 1;
```

Database altered.

```
SYS@orcl> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status, b.archived, b.first_change#,  
b.next_change# from v$logfile a, v$log b where a.group# = b.group# order by 1,2;
```

- unarchived를 빼고 clear 복구.

=====

<<시나리오 22.1>> 22번 시나리오를 clear unarchived로 복구

```
SYS@orcl> alter system switch logfile;
```

```
System altered.
```

```
SYS@orcl> /
```

```
System altered.
```

```
SYS@orcl> /
```

```
System altered.
```

```
SYS@orcl> /
```

```
System altered.
```

```
SYS@orcl> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status, b.archived, b.first_change#,
b.next_change# from v$logfile a, v$log b where a.group# = b.group# order by 1,2;
```

GROUP#	SEQUENCE#	MEMBER	MB	STATUS	ARC	FIRST_CHANGE#	NEXT_CHANGE#
1	13	/u01/app/oracle/oradata/ORCL/redo01.log	10	CURRENT	NO	3114935	1.8447E+19
2	11	/u01/app/oracle/oradata/ORCL/redo02.log	10	INACTIVE	YES	3114928	3114931

GROUP#	SEQUENCE#	MEMBER	MB	STATUS	ARC	FIRST_CHANGE#	NEXT_CHANGE#
1	13	/u01/app/oracle/oradata/ORCL/redo01.log	10	CURRENT	NO	3114935	1.8447E+19
2	11	/u01/app/oracle/oradata/ORCL/redo02.log	10	INACTIVE	YES	3114928	3114931

```
          3          12
/u01/app/oracle/oradata/ORCL/redo03.log
      10 INACTIVE      YES      3114931      3114935
```

장애 발생

```
SYS@orcl> ! rm /u01/app/oracle/oradata/ORCL/redo01.log
```

```
SYS@orcl> ! ls /u01/app/oracle/oradata/ORCL/redo01.log
```

```
ls: cannot access /u01/app/oracle/oradata/ORCL/redo01.log: No such file or directory
```

정상종료

```
SYS@orcl> shutdown immediate
```

```
Database closed.
```

```
Database dismounted.
```

```
ORACLE instance shut down.
```

오류

```
SYS@orcl> startup
```

```
ORACLE instance started.
```

```
Total System Global Area 1543500144 bytes
```

```
Fixed Size                  8896880 bytes
```

```
Variable Size              905969664 bytes
```

```
Database Buffers          620756992 bytes
```

```
Redo Buffers               7876608 bytes
```

```
Database mounted.
```

```
ORA-03113: end-of-file on communication channel
```

```
Process ID: 29689
```

```
Session ID: 1 Serial number: 8953
```

```
-- Alert log file
```

```
2025-05-14T00:37:37.045028+09:00
```

```
Errors in file /u01/app/oracle/diag/rdbms/orcl/orcl/trace/orcl_lgwr_29649.trc:
```

```
ORA-00313: open failed for members of log group 1 of thread 1
```

ORA-00312: online log 1 thread 1: '/u01/app/oracle/oradata/ORCL/redo01.log'

ORA-27037: unable to obtain file status

Linux-x86_64 Error: 2: No such file or directory

Additional information: 7

복구

SYS@orcl> conn /as sysdba

Connected to an idle instance.

SYS@orcl> startup mount

ORACLE instance started.

Total System Global Area 1543500144 bytes

Fixed Size 8896880 bytes

Variable Size 905969664 bytes

Database Buffers 620756992 bytes

Redo Buffers 7876608 bytes

Database mounted.

SYS@orcl> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status, b.archived, b.first_change#,
b.next_change# from v\$logfile a, v\$log b where a.group# = b.group# order by 1,2;

GROUP#	SEQUENCE#	MEMBER	MB	STATUS	ARC	FIRST_CHANGE#	NEXT_CHANGE#
1	13	/u01/app/oracle/oradata/ORCL/redo01.log	10	CURRENT	NO	3114935	1.8447E+19
2	11	/u01/app/oracle/oradata/ORCL/redo02.log	10	INACTIVE	YES	3114928	3114931

GROUP#	SEQUENCE#	MEMBER	MB	STATUS	ARC	FIRST_CHANGE#	NEXT_CHANGE#
1	13	/u01/app/oracle/oradata/ORCL/redo01.log	10	CURRENT	NO	3114935	1.8447E+19
2	11	/u01/app/oracle/oradata/ORCL/redo02.log	10	INACTIVE	YES	3114928	3114931


```

-----
          3          12
/u01/app/oracle/oradata/ORCL/redo03.log
          10 INACTIVE          YES          3114931          3114935

```

```

SYS@orcl> alter database clear unarchived logfile group 1;
Database altered.

```

```

SYS@orcl> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status, b.archived, b.first_change#,
b.next_change# from v$logfile a, v$log b where a.group# = b.group# order by 1,2;

```

```

GROUP# SEQUENCE#
-----
MEMBER
-----
          MB STATUS          ARC FIRST_CHANGE# NEXT_CHANGE#
-----
          1          0
/u01/app/oracle/oradata/ORCL/redo01.log
          10 UNUSED          YES          3114935          3115055

          2          14
/u01/app/oracle/oradata/ORCL/redo02.log
          10 CURRENT          NO          3115055 1.8447E+19

```

```

GROUP# SEQUENCE#
-----
MEMBER
-----
          MB STATUS          ARC FIRST_CHANGE# NEXT_CHANGE#
-----

          3          12
/u01/app/oracle/oradata/ORCL/redo03.log
          10 INACTIVE          YES          3114931          3114935

```

```

SYS@orcl> alter database open;
Database altered.

```

```

# Online backup( sh onbkp.sh 수행)

```

<<시나리오23>> DB 정상적인 종료 후에 controlfile 손상되었을 경우 복구

```
SYS@orcl> show parameter control_file
```

NAME	TYPE	VALUE
control_file_record_keep_time	integer	7
control_files	string	/u01/app/oracle/oradata/ORCL

```
SYS@orcl> shutdown immediate
```

Database closed.

Database dismounted.

ORACLE instance shut down.

장애 발생

```
SYS@orcl> ! rm /u01/app/oracle/oradata/ORCL/control01.ctl
```

```
SYS@orcl> ! ls /u01/app/oracle/oradata/ORCL/control01.ctl
```

ls: cannot access /u01/app/oracle/oradata/ORCL/control01.ctl: No such file or directory

```
SYS@orcl> startup
```

ORACLE instance started.

Total System Global Area 1140849904 bytes

Fixed Size 8895728 bytes

Variable Size 788529152 bytes

Database Buffers 335544320 bytes

Redo Buffers 7880704 bytes

ORA-00205: error in identifying control file, check alert log for more info

-- Alert log file

```
ALTER DATABASE MOUNT
```

2025-05-14T00:45:09.587904+09:00

ORA-00210: cannot open the specified control file

ORA-00202: control file: '/u01/app/oracle/oradata/ORCL/control01.ctl'

ORA-27037: unable to obtain file status

Linux-x86_64 Error: 2: No such file or directory

Additional information: 7

```
SYS@orcl> select status from v$instance;
```

```
STATUS
```

```
-----
```

```
STARTED
```

```
# Recovery
```

1. 최근 컨트롤파일 백업본을 찾아 restore

```
SYS@orcl> ! ls -l /home/oracle/backup/open_bkp/control01.ctl
```

```
-rw-r-----. 1 oracle dba 10829824 May 14 00:43 /home/oracle/backup/open_bkp/control01.ctl
```

```
SYS@orcl> ! cp -v /home/oracle/backup/open_bkp/control01.ctl /u01/app/oracle/oradata/ORCL/  
'/home/oracle/backup/open_bkp/control01.ctl' -> '/u01/app/oracle/oradata/ORCL/control01.ctl'
```

2. DB mount

```
SYS@orcl> alter database mount;
```

```
Database altered.
```

3. 백업 이후에 변경정보(리두) 적용

```
SYS@orcl> recover database using backup controlfile
```

```
ORA-00279: change 3300451 generated at 02/26/2025 14:41:16 needed for thread 1
```

```
ORA-00289: suggestion : /home/oracle/arch2/arch_1_11_1194089464.arc
```

```
ORA-00280: change 3300451 for thread 1 is in sequence #11
```

```
Specify log: {<RET>=suggested | filename | AUTO | CANCEL}
```

```
auto
```

```
ORA-00279: change 3300467 generated at 02/26/2025 15:18:40 needed for thread 1
```

```
ORA-00289: suggestion : /home/oracle/arch2/arch_1_12_1194089464.arc
```

```
ORA-00280: change 3300467 for thread 1 is in sequence #12
```

```
ORA-00278: log file '/home/oracle/arch2/arch_1_11_1194089464.arc' no longer needed for this recovery
```

```
ORA-00308: cannot open archived log '/home/oracle/arch2/arch_1_12_1194089464.arc'
```

```
ORA-27037: unable to obtain file status
```

```
Linux-x86_64 Error: 2: No such file or directory
```

```
Additional information: 7
```

- 온라인백업시 데이터파일 백업, 컨트롤 파일 백업 이후 아카이브를 백업하여 컨트롤 파일은 이전 리두정보를 찾음
- 리두정보는 2그룹이 있고 어떤그룹이 찾고 있는 그룹인지 사용자는 알수없음으로 얼럿로그에서 찾아야 함

-- Alert log file

alter database backup controlfile to '/home/oracle/backup/open_bkp/control01.ctl'

Completed: alter database backup controlfile to '/home/oracle/backup/open_bkp/control01.ctl'

2025-05-14T00:43:16.076363+09:00

ALTER SYSTEM ARCHIVE LOG

2025-05-14T00:43:16.084154+09:00

Thread 1 advanced to log sequence 15 (LGWR switch)

Current log# 1 seq# 15 mem# 0: /u01/app/oracle/oradata/ORCL/redo01.log

SYS@orcl> recover database using backup controlfile

ORA-00279: change 3115903 generated at 05/14/2025 00:43:16 needed for thread 1

ORA-00289: suggestion : /home/oracle/arch2/arch_1_15_1201045123.arc

ORA-00280: change 3115903 for thread 1 is in sequence #15

Specify log: {<RET>=suggested | filename | AUTO | CANCEL}

/u01/app/oracle/oradata/ORCL/redo01.log

Log applied.

Media recovery complete.

4. DB open (resetlogs)

SYS@orcl> alter database open resetlogs;

Database altered.

- control file 재생성도 가능

Online backup(sh onbkp.sh 수행)

<시나리오24>> DB 정상적인 종료 후에 controlfile 손상되었을 경우 복구(컨트롤파일 재생성)

```
SYS@orcl> show parameter control_file;
```

NAME	TYPE	VALUE
control_file_record_keep_time	integer	7
control_files	string	/u01/app/oracle/oradata/ORCL

```
SYS@orcl> shutdown immediate
```

Database closed.

Database dismounted.

ORACLE instance shut down.

장애 발생

```
SYS@orcl> ! rm /u01/app/oracle/oradata/ORCL/control01.ctl
```

```
SYS@orcl> ! ls /u01/app/oracle/oradata/ORCL/control01.ctl
```

ls: cannot access /u01/app/oracle/oradata/orcl/control01.ctl: No such file or directory

```
SYS@orcl> startup
```

ORACLE instance started.

Total System Global Area 1543500144 bytes

Fixed Size 8896880 bytes

Variable Size 905969664 bytes

Database Buffers 620756992 bytes

Redo Buffers 7876608 bytes

ORA-00205: error in identifying control file, check alert log for more info

-- Alert log file

2025-05-14T01:01:40.812792+09:00

Errors in file /u01/app/oracle/diag/rdbms/orcl/orcl/trace/orcl_mz00_31901.trc:

ORA-00202: control file: '/u01/app/oracle/oradata/ORCL/control01.ctl'

ORA-27037: unable to obtain file status

Linux-x86_64 Error: 2: No such file or directory

Additional information: 7

```
SYS@orcl> select status from v$instance;
```

```
STATUS
```

```
-----
```

```
STARTED
```

```
# Recovery
```

1. 최근 컨트롤파일 백업본을 찾아 restore

```
SYS@orcl> ! ls -l /home/oracle/backup/open_bkp/control01.ctl
```

```
-rw-r-----. 1 oracle dba 10829824 May 14 00:57 /home/oracle/backup/open_bkp/control01.ctl
```

```
SYS@orcl> ! cp -v /home/oracle/backup/open_bkp/control01.ctl /u01/app/oracle/oradata/ORCL/  
'/home/oracle/backup/open_bkp/control01.ctl' -> '/u01/app/oracle/oradata/ORCL/control01.ctl'
```

2. DB mount

```
SYS@orcl> alter database mount;
```

```
Database altered.
```

```
SYS@orcl> select status from v$instance;
```

```
STATUS
```

```
-----
```

```
MOUNTED
```

3. 컨트롤파일 trace 파일 생성

```
SYS@orcl> alter database backup controlfile to trace as '/home/oracle/create_control.txt';
```

```
Database altered.
```

4. DB 종료

```
SYS@orcl> shutdown abort
```

```
ORACLE instance shut down.
```

5. 컨트롤파일 재생성

데이터파일 문제 없음, 리두 문제 없음 -> noresetlogs

```
SYS@orcl> !vi cre_con.sql
```

```
STARTUP NOMOUNT
```

```
CREATE CONTROLFILE REUSE DATABASE "orcl" NORESETLOGS ARCHIVELOG
```

```
    MAXLOGFILES 16
```

```
    MAXLOGMEMBERS 3
```

```
    MAXDATAFILES 100
```

```
    MAXINSTANCES 8
```

```
    MAXLOGHISTORY 292
```

```
LOGFILE
```

```
    GROUP 1 '/u01/app/oracle/oradata/ORCL/redo01.log' SIZE 10M BLOCKSIZE 512,
```

```
    GROUP 2 '/u01/app/oracle/oradata/ORCL/redo02.log' SIZE 10M BLOCKSIZE 512,
```

```
    GROUP 3 '/u01/app/oracle/oradata/ORCL/redo03.log' SIZE 10M BLOCKSIZE 512
```

```
-- STANDBY LOGFILE
```

```
DATAFILE
```

```
    '/u01/app/oracle/oradata/ORCL/system01.dbf',
```

```
    '/u01/app/oracle/oradata/ORCL/sysaux01.dbf',
```

```
    '/u01/app/oracle/oradata/ORCL/undotbs01.dbf',
```

```
    '/u01/app/oracle/oradata/ORCL/users01.dbf'
```

```
CHARACTER SET AL32UTF8
```

```
;
```

```
ALTER DATABASE OPEN;
```

```
ALTER TABLESPACE TEMP ADD TEMPFILE '/u01/app/oracle/oradata/orcl/temp01.dbf' REUSE;
```

```
SYS@orcl> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status, b.archived, b.first_change#,  
b.next_change# from v$logfile a, v$log b where a.group# = b.group# order by 1,2;
```

```
SYS@orcl> select sequence#, name, FIRST_CHANGE#, NEXT_CHANGE# from v$archived_log;
```

<<시나리오25>> DB 비정상적인 종료 후에 controlfile 손상되었을 경우 복구(컨트롤파일 재생성)

```
SYS@orcl> show parameter control_file;
```

NAME	TYPE	VALUE
control_file_record_keep_time	integer	7
control_files	string	/u01/app/oracle/oradata/ORCL

```
SYS@orcl> shutdown abort  
ORACLE instance shut down.
```

장애 발생

```
SYS@orcl> ! rm /u01/app/oracle/oradata/ORCL/control01.ctl
```

```
SYS@orcl> ! ls /u01/app/oracle/oradata/ORCL/control01.ctl  
ls: cannot access /u01/app/oracle/oradata/orcl/control01.ctl: No such file or directory
```

```
SYS@orcl> startup  
ORACLE instance started.
```

```
Total System Global Area 1543500144 bytes  
Fixed Size 8896880 bytes  
Variable Size 905969664 bytes  
Database Buffers 620756992 bytes  
Redo Buffers 7876608 bytes  
ORA-00205: error in identifying control file, check alert log for more info
```

```
-- Alert log file  
ALTER DATABASE MOUNT  
2025-05-14T01:14:18.535525+09:00  
ORA-00210: cannot open the specified control file  
ORA-00202: control file: '/u01/app/oracle/oradata/ORCL/control01.ctl'  
ORA-27037: unable to obtain file status  
Linux-x86_64 Error: 2: No such file or directory  
Additional information: 7
```



```
SYS@orcl> select status from v$instance;
```

```
STATUS
```

```
-----
```

```
STARTED
```

```
# Recovery
```

1. 최근 컨트롤파일 백업본을 찾아 restore

```
SYS@orcl> shut abort
```

```
ORACLE instance shut down.
```

```
SYS@orcl> startup nomount
```

```
SYS@orcl> ! ls -l /home/oracle/backup/open_bkp/control01.ctl
```

```
-rw-r-----. 1 oracle dba 10928128 May 14 01:11 /home/oracle/backup/open_bkp/control01.ctl
```

```
SYS@orcl> ! cp -v /home/oracle/backup/open_bkp/control01.ctl /u01/app/oracle/oradata/ORCL/  
'/home/oracle/backup/open_bkp/control01.ctl' -> '/u01/app/oracle/oradata/ORCL/control01.ctl'
```

2. DB mount

```
SYS@orcl> alter database mount;
```

```
Database altered.
```

```
SYS@orcl> select status from v$instance;
```

```
STATUS
```

```
-----
```

```
MOUNTED
```

3. 컨트롤파일 trace 파일 생성

```
SYS@orcl> alter database backup controlfile to trace as '/home/oracle/cre_control.txt';
```

```
Database altered.
```

4. DB 종료

```
SYS@orcl> shutdown abort
ORACLE instance shut down.
```

5. 컨트롤파일 재생성

데이터파일 문제 없음, 리두 문제 없음 -> noresetlogs

```
SYS@orcl> !vi cre_con1.sql
STARTUP NOMOUNT
CREATE CONTROLFILE REUSE DATABASE "orcl" NORESETLOGS  ARCHIVELOG
    MAXLOGFILES 16
    MAXLOGMEMBERS 3
    MAXDATAFILES 100
    MAXINSTANCES 8
    MAXLOGHISTORY 292
LOGFILE
    GROUP 1 '/u01/app/oracle/oradata/ORCL/redo01.log'  SIZE 10M BLOCKSIZE 512,
    GROUP 2 '/u01/app/oracle/oradata/ORCL/redo02.log'  SIZE 10M BLOCKSIZE 512,
    GROUP 3 '/u01/app/oracle/oradata/ORCL/redo03.log'  SIZE 10M BLOCKSIZE 512
-- STANDBY LOGFILE
DATAFILE
    '/u01/app/oracle/oradata/ORCL/system01.dbf',
    '/u01/app/oracle/oradata/ORCL/sysaux01.dbf',
    '/u01/app/oracle/oradata/ORCL/undotbs01.dbf',
    '/u01/app/oracle/oradata/ORCL/users01.dbf'

CHARACTER SET AL32UTF8
;
RECOVER DATABASE
ALTER DATABASE OPEN;
```

```
SYS@orcl> @cre_con1.sql
ORACLE instance started.
```

```
Total System Global Area 1140849904 bytes
Fixed Size                  8895728 bytes
Variable Size               788529152 bytes
Database Buffers           335544320 bytes
Redo Buffers                7880704 bytes
```

Control file created.

Media recovery complete.

Database altered.

```
SYS@orcl> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status, b.archived, b.first_change#,  
b.next_change# from v$logfile a, v$log b where a.group# = b.group# order by 1,2;
```

```
SYS@orcl> select sequence#, name, FIRST_CHANGE#, NEXT_CHANGE# from v$archived_log;
```